

Chapter 6

CMOS Operational Amplifiers

The operational amplifier, which has become one of the most versatile and important building blocks in analog circuit design, is introduced in this chapter. The operational amplifier (op amp) fits into the scheme of Table 1.1-2 as an example of a complex circuit. The unbuffered operational amplifiers developed in this chapter might be better described as operational-transconductance amplifiers since the output resistance typically will be very high (hence the term “unbuffered”). The term “op amp” has become accepted for such circuits, so it will be used throughout this text. The terms “unbuffered” and “buffered” will be used to distinguish between high output resistance (operational-transconductance amplifiers or OTAs) and low output resistance amplifiers (voltage operational amplifiers). Chapter 7 will examine op amps that have low output resistance (buffered op amps).

Operational amplifiers are amplifiers (controlled sources) that have sufficiently high forward gain so that when negative feedback is applied, the closed-loop transfer function is practically independent of the gain of the op amp (see Fig. 5.6-1). This principle has been exploited to develop many useful analog circuits and systems. The primary requirement of an op amp is to have an open-loop gain that is sufficiently large to implement the negative feedback concept. Most of the amplifiers in Chapter 5 do not have a large enough gain. Consequently, most CMOS op amps use two or more stages of gain. One of the most popular op amps is a two-stage op amp. We will carefully examine the performance of this type of op amp for several reasons. The first is because it is a simple yet robust implementation of an op amp and second, it can be used as the starting point for the development of other types of op amps.

The two-stage op amp will be used to introduce the important concept of compensation. The goal of compensation is to maintain stability when negative feedback is applied around the op amp. An understanding of compensation, along with the previous concepts, provides the necessary design relationships to formulate a design approach for the two-stage op amp. In addition to the two-stage op amp, this chapter will examine the folded-cascode op amp. This amplifier was developed in order to improve the power-supply rejection ratio performance of the two-stage op amp. The folded-cascode op amp is also an example of a self-compensated op amp.

Simulation, measurement, and macromodeling of op amp performance will be the final subjects of this chapter. Simulation is necessary to verify and refine the design. Experimental measurements are necessary to verify the performance of the op amp with the original design specifications. Typically, the techniques applicable to simulation are also suitable for experimental measurement. Finally, macromodeling captures the desired performance of the op amp without modeling every component of the op amp.

6.1 DESIGN OF CMOS OP AMPS

Figure 6.1-1 shows a block diagram that represents the important aspects of an op amp. CMOS op amps are very similar in architecture to their bipolar counterparts. The differential-transconductance stage introduced in Section 5.2 forms the input of the op amp and sometimes provides the differential-to-single-ended conversion. Normally, a good portion of the overall gain is provided by the differential-input stage, which improves noise and offset performance. The second stage is typically an inverter similar to that introduced in Section 5.1. If the differential-input stage does not perform the differential-to-single-ended conversion, then it is accomplished in the second-stage inverter. If the op amp must drive a low-resistance load, the second stage must be followed by a buffer stage whose objective is to lower the output resistance and maintain a large signal swing. Bias circuits are provided to establish the proper operating point for each transistor in its quiescent state. As noted in the introduction, compensation is required to achieve stable closed-loop performance. Section 6.2 will address this important topic.

Ideal Op Amp

Ideally, an op amp has infinite differential-voltage gain, infinite input resistance, and zero output resistance. In reality, an op amp only approaches these values. For most applications where unbuffered CMOS op amps are used, an open-loop gain of 2000 or more is usually sufficient. The symbol for an op amp is shown in Fig. 6.1-2, where, in the nonideal case, the output voltage v_{OUT} can be expressed as

$$v_{OUT} = A_v (v_1 - v_2) \quad (6.1-1)$$

A_v is used to designate the open-loop differential-voltage gain. v_1 and v_2 are the input voltages applied to the noninverting and inverting terminals, respectively. The symbol in Fig. 6.1-2 also shows the power-supply connections of V_{DD} and V_{SS} . Generally, these connections are not shown but the designer must remember that they are an integral part of the op amp.

If the gain of the op amp is large enough, the input port of the op amp becomes a *null port* when negative feedback is applied. A null port (or *nullor*) is a pair of terminals to a network

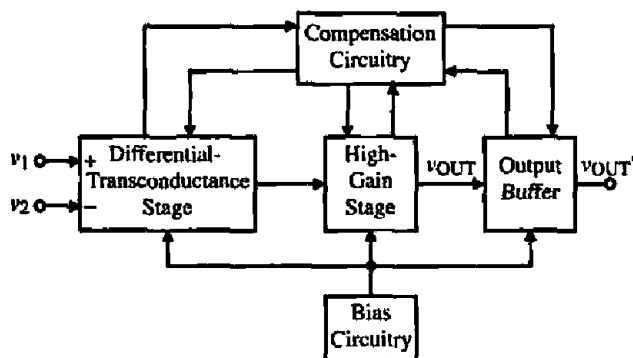


Figure 6.1-1 Block diagram of a general two-stage op amp.

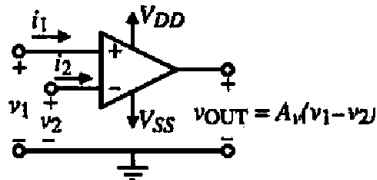


Figure 6.1-2 Symbol for an operational amplifier.

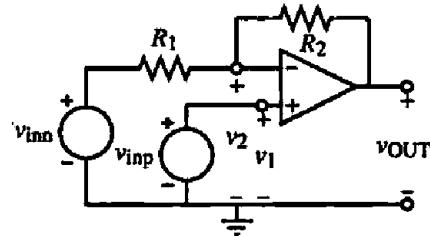


Figure 6.1-3 General configuration of the op amp as a voltage amplifier.

where the voltage across the terminals is zero and the current flowing into or out of the terminals is also zero [1]. In terms of Fig. 6.1-2, if we define

$$v_t = v_1 - v_2 \quad (6.1-2)$$

and

$$i_t = i_1 = -i_2 \quad (6.1-3)$$

then

$$v_t = i_t = 0 \quad (6.1-4)$$

This concept permits the analysis of op amp circuits with negative feedback to be very simple. This will be illustrated shortly.

Figure 6.1-3 shows the typical implementation of a voltage amplifier using an op amp. Returning the output through R_2 to the inverting input provides the negative feedback path. The input may be applied at the positive or negative inputs. If only v_{inp} is applied ($v_{inn} = 0$), the voltage amplifier is called *noninverting*. If only v_{inn} is applied ($v_{inp} = 0$), the voltage amplifier is called *inverting*.

SIMPLIFIED ANALYSIS OF AN OP AMP CIRCUIT

The circuit shown in Fig. 6.1-4 is an inverting voltage amplifier using an op amp. Find the voltage-transfer function, v_{out}/v_{in} .

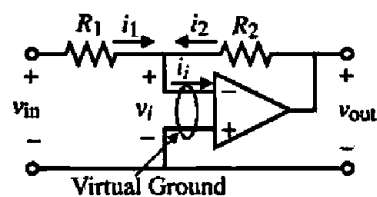


Figure 6.1-4 Inverting voltage amplifier using an op amp.

Solution

If the differential-voltage gain, A_v , is large enough, then the negative feedback path through R_2 will cause the voltage v_i and the current i_i shown on Fig. 6.1-4 to both be zero. Note that the null port becomes the familiar *virtual ground* if one of the op amp input terminals is on ground. If this is the case, then we can write that

$$i_1 = \frac{v_{in}}{R_1}$$

and

$$i_2 = \frac{v_{out}}{R_2}$$

Since, $i_i = 0$, then $i_1 + i_2 = 0$, giving the desired result as

$$\frac{v_{out}}{v_{in}} = -\frac{R_2}{R_1}$$

Characterization of Op Amps

In practice, the operational amplifier only approaches the ideal infinite-gain voltage amplifier. Some of its other nonideal characteristics are illustrated in Fig. 6.1-5. The *finite differential-input impedance* is modeled by R_{id} and C_{id} . The *output resistance* is modeled by R_{out} . The *common-mode input resistances* are given as resistors of R_{icm} connected from each of the inputs to ground. V_{OS} is the *input-offset voltage* necessary to make the output voltage zero if both of the inputs of the op amp are grounded. I_{OS} (not shown) is the *input-offset current*, which is necessary to make the output voltage zero if the op amp is driven from two identical current sources. Therefore, I_{OS} is defined as the magnitude of the difference between the two *input-bias currents* I_{B1} and I_{B2} . Since the bias currents for a CMOS op amp are approximately zero, the offset current is also zero. The *common-mode rejection ratio* (CMRR) is modeled

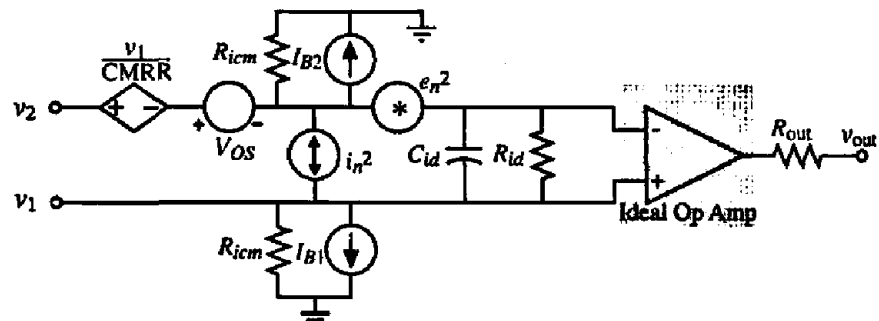


Figure 6.1-5 A model for a nonideal op amp showing some of the nonideal linear characteristics.

by the voltage-controlled voltage source indicated as v_1/CMRR . This source approximately models the effects of the common-mode input signal on the op amp. The two sources designated as e_n^2 and i_n^2 are used to model the op amp noise. These are *rms voltage- and current-noise sources* with units of mean-square volts and mean-square amperes, respectively. These noise sources have no polarity and are always assumed to add.

Not all of the nonideal characteristics of the op amp are shown in Fig. 6.1-5. Other pertinent characteristics of the op amp will now be defined. The output voltage of Fig. 6.1-2 can be defined as

$$V_{\text{out}}(s) = A_v(s) [V_1(s) - V_2(s)] \pm A_c(s) \left(\frac{V_1(s) + V_2(s)}{2} \right) \quad (6.1-5)$$

where the first term on the right is the differential portion of $V_{\text{out}}(s)$ and the second term is the common-mode portion of $V_{\text{out}}(s)$. The *differential-frequency response* is given as $A_v(s)$ while the *common-mode frequency response* is given as $A_c(s)$. A typical differential-frequency response of an op amp is given as

$$A_v(s) = \frac{A_{v0}}{\left(\frac{s}{p_1} - 1 \right) \left(\frac{s}{p_2} - 1 \right) \left(\frac{s}{p_3} - 1 \right) \cdots} \quad (6.1-6)$$

where $p_1, p_2, \dots$ are poles of the operational amplifier open-loop transfer function. In general, a pole designated as p_i can be expressed as

$$p_i = -\omega_i \quad (6.1-7)$$

where ω_i is the reciprocal time constant or break-frequency of the pole p_i . While the operational amplifier may have zeros, they will be ignored at the present time. A_{v0} or $A_v(0)$ is the gain of the op amp as the frequency approaches zero. Figure 6.1-6 shows a typical frequency response of the magnitude of $A_v(s)$. In this case we see that ω_1 is much lower than the rest of the break-frequencies, causing ω_1 to be the dominant influence in the frequency response. The

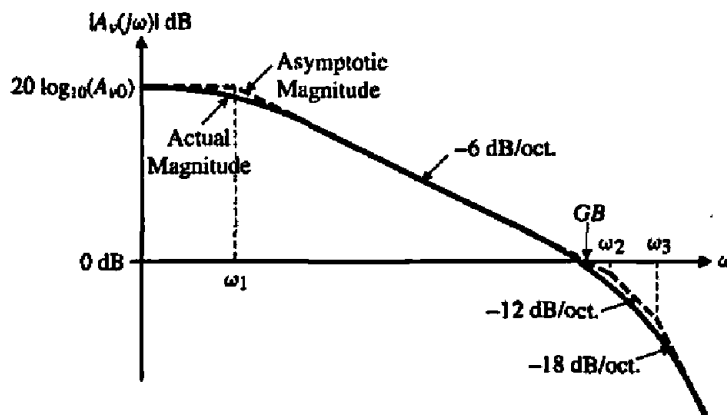


Figure 6.1-6 Typical frequency response of the magnitude of $A_v(j\omega)$ for an op amp.

frequency where the -6 dB/oct. slope from the dominant pole intersects with the 0 dB axis is designated as the *unity-gain bandwidth*, abbreviated *GB*, of the op amp. Even if the next higher order poles are smaller than *GB*, we shall still continue to use the unity-gain bandwidth as defined above.

Other nonideal characteristics of the op amp not defined by Fig. 6.1-5 include the *power-supply rejection ratio*, PSRR. The PSRR is defined as the product of the ratio of the change in supply voltage to the change in output voltage of the op amp caused by the change in the power supply and the open-loop gain of the op amp. Thus,

$$\text{PSRR} = \frac{\Delta V_{DD}}{\Delta V_{OUT}} A_v(s) = \frac{V_o/V_{in}(V_{dd} = 0)}{V_o/V_{dd}(V_{in} = 0)} \quad (6.1-8)$$

An ideal op amp would have an infinite PSRR. The reader is advised that both this definition for PSRR and its inverse will be found in the literature. The *common-mode input range* is the voltage range over which the input common-mode signal can vary. Typically, this range is 1–2 V less than V_{DD} and 1–2 V more than V_{SS} .

The output of the op amp has several important limits, one of which is the maximum output current sourcing and sinking capability. There is a limited range over which the output voltage can swing while still maintaining high-gain characteristics. The output also has a voltage rate limit called *slew rate*. The slew rate is generally determined by the maximum current available to charge or discharge a capacitance. Normally, slew rate is not limited by the output, but by the current sourcing/sinking capability of the first stage. The last characteristic of importance in analog sampled-data circuit applications is the *settling time*. This is the time needed for the output of the op amp to reach a final value (to within a predetermined tolerance) when excited by a small signal. This is not to be confused with slew rate, which is a large-signal phenomenon. Many times, the output response of an op amp is a combination of both large- and small-signal characteristics. Small-signal settling time can be completely determined from the location of the poles and zeros in the small-signal equivalent circuit, whereas slew rate is determined from the large-signal conditions of the circuit.

The importance of the settling time to analog sampled-data circuits is illustrated by Fig. 6.1-7. It is necessary to wait until the amplifier has settled to within a few tenths of a percent of its final value in order to avoid errors in the accuracy of processing analog signals. A longer settling time implies that the rate of processing analog signals must be reduced.

Fortunately, the CMOS op amp does not suffer from all of the nonideal characteristics previously discussed. Because of the extremely high input resistance of the MOS devices, both R_{id} and I_{OS} (or I_{B1} and I_{B2}) are of no importance. A typical value of R_{id} is in the range of $10^{14} \Omega$. Also, R_{icm} is extremely large and can be ignored. If an op amp is used in the configuration of Fig. 6.1-3 with the noninverting terminal on ac ground, then all common-mode characteristics are unimportant.

Classification of Op Amps

In order to understand the design of CMOS op amps it is worthwhile to examine their classification and categorization. It is encouraging that op amps that we are totally unfamiliar with at this point can be implemented using the blocks of the previous two chapters. Table 6.1-1 gives a hierarchy of CMOS op amps that is applicable to nearly all CMOS op amps that will be presented in this chapter and the next. We see that the differential amplifier is almost ubiq-

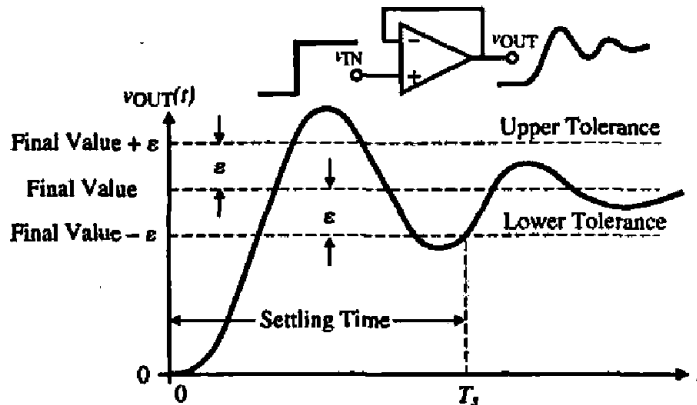


Figure 6.1-7 Transient response of an op amp with negative feedback illustrating settling time T_s . ϵ is the tolerance to the final value used to define the settling time.

uitous in its use as the input stage. We will study several op amps in the next chapter that use a modified form of the differential amplifier of Section 5.2 but by and large the differential amplifier is the input stage of choice for most op amps.

As we have shown previously, amplifiers generally consist of a cascade of voltage-to-current or current-to-voltage converting stages. A voltage-to-current stage is called a *transconductance stage* and a current-to-voltage stage is called the *load stage*. In some cases, it is easier to think of a current-to-current stage, but eventually current will be converted back to voltage.

Based on the categorization in Table 6.1-1, there are two major op amp architectures that we will study in this chapter. The first is the two-stage op amp. It consists of a cascade of $V \rightarrow I$ and $I \rightarrow V$ stages and is shown in Fig. 6.1-8. The first stage consists of a differential amplifier converting the differential input voltage to differential currents. These differential currents are applied to a current-mirror load recovering the differential voltage. This of course is nothing more than the differential voltage amplifier of Fig. 5.2-5 or 5.2-7. The second stage consists of a common-source MOSFET converting the second-stage input voltage to current. This transistor is loaded by a current-sink load, which converts the current to voltage at the output. The

TABLE 6.1-1 Categorization of CMOS Op Amps

Conversion	Hierarchy		
Voltage to current	Classic differential amplifier	Modified differential amplifier	First voltage stage
Current to voltage	Differential-to-single-ended load (current mirror)	Source/sink current loads MOS diode load	
Voltage to current	Transconductance grounded gate	Transconductance grounded source	Current stage
Current to voltage	Class A (source or sink load)	Class B (push-pull)	Second voltage stage

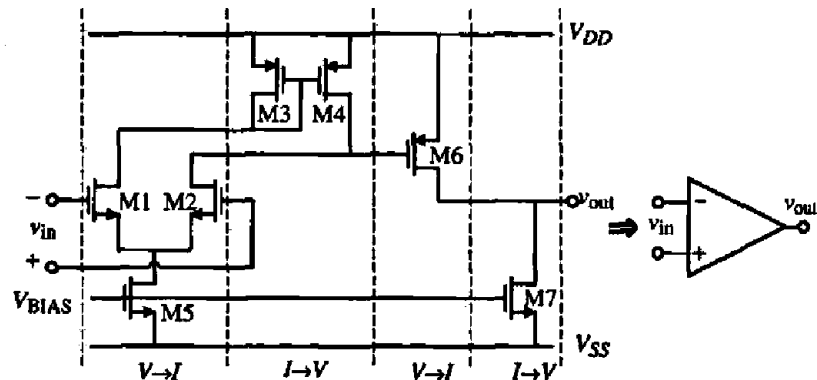


Figure 6.1-8 Classical two-stage CMOS op amp broken into voltage-to-current and current-to-voltage stages.

second stage is also nothing more than the current-sink inverter of Fig. 5.1-7. This two-stage op amp is so widely used that we will call it the *classical two-stage op amp*; it has both MOSFET and BJT versions.

A second architecture that results is shown in Fig. 6.1-9. This architecture is commonly called the *folded-cascode op amp*. This architecture was developed in part to improve the input common-mode range and the power-supply rejection of the two-stage op amp. In this particular op amp, it is probably more efficient to consider it as the cascade of a differential-transconductance stage with a current stage followed by a cascode current-mirror load. One of the advantages of the folded-cascode op amp is that it has a push-pull output. That is, the op amp can actively sink or source current from the load. The output stage of the previous two-stage op amp is Class A, which means that either its sinking or sourcing capability is fixed.

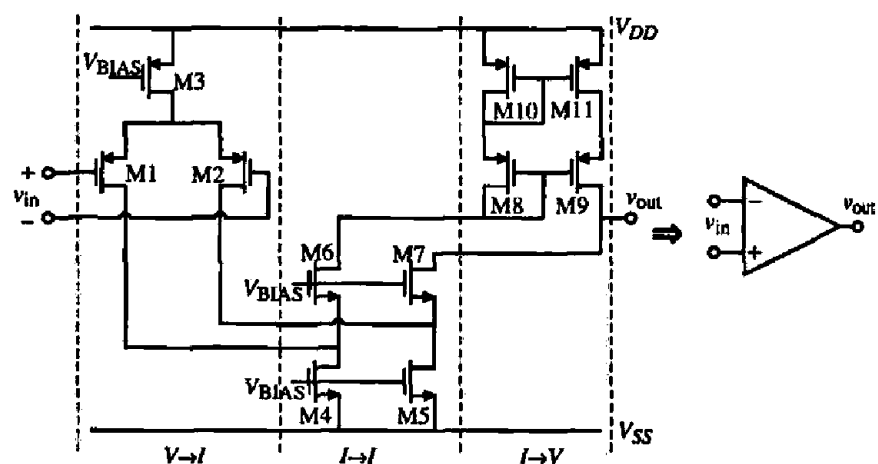


Figure 6.1-9 Folded-cascode op amp broken into stages.

Slight modifications in the two op amps of Figs. 6.1-8 and 6.1-9 result in many possible other forms (see Problems 6.1-5 and 6.1-6). However, for the purposes of space and simplicity we will restrict our present considerations to these two CMOS op amps.

Design of Op Amps

The design of an op amp can be divided into two distinct design-related activities that are for the most part independent of one another. The first of these activities involves choosing or creating the basic structure of the op amp. A diagram that describes the interconnection of all of the transistors results. In most cases, this structure does not change throughout the remaining portion of the design, but sometimes certain characteristics of the chosen design must be changed by modifying the structure.

Once the structure has been selected, the designer must select dc currents and begin to size the transistors and design the compensation circuit. Most of the work involved in completing a design is associated with this, the second activity of the design process. Devices must be properly scaled in order to meet all of the ac and dc requirements imposed on the op amp. Computer circuit simulations, based on hand calculations, are used extensively to aid the designer in this phase.

Before the actual design of an op amp can begin, though, one must set out all of the requirements and boundary conditions that will be used to guide the design. The following list describes many of the items that must be considered.

Boundary conditions:

1. Process specification (V_T , K' , C_{ox} , etc.)
2. Supply voltage and range
3. Supply current and range
4. Operating temperature and range

Requirements:

1. Gain
2. Gain bandwidth
3. Settling time
4. Slew rate
5. Input common-mode range, ICMR
6. Common-mode rejection ratio, CMRR
7. Power-supply rejection ratio, PSRR
8. Output-voltage swing
9. Output resistance
10. Offset
11. Noise
12. Layout area

The typical specifications for an unbuffered CMOS op amp are listed in Table 6.1-2.

TABLE 6.1-2 Specifications for a Typical Unbuffered CMOS Op Amp

Boundary Conditions	Requirement
Process specification	See Tables 3.1-1, 3.1-2, and 3.2-1
Supply voltage	$\pm 2.5 \text{ V} \pm 10\%$
Supply current	$100 \mu\text{A}$
Temperature range	$0\text{--}70^\circ\text{C}$
Specifications	
Gain	$\geq 70 \text{ dB}$
Gain bandwidth	$\geq 5 \text{ MHz}$
Settling time	$\leq 1 \mu\text{s}$
Slew rate	$\geq 5 \text{ V}/\mu\text{s}$
ICMR	$\geq \pm 1.5 \text{ V}$
CMRR	$\geq 60 \text{ dB}$
PSRR	$\geq 60 \text{ dB}$
Output swing	$\geq \pm 1.5 \text{ V}$
Output resistance	N/A, capacitive load only
Offset	$\leq \pm 10 \text{ mV}$
Noise	$\leq 100 \text{ nV}/\sqrt{\text{Hz}}$ at 1 kHz
Layout area	$\leq 5000 \times (\text{minimum channel length})^2$

The block diagram of Fig. 6.1-1 is useful for guiding the CMOS op amp design process. The compensation method has a large influence on the design of each block. Two basic methods of compensation are suggested by the opposite parallel paths into the compensation block of Fig. 6.1-1. These two methods, feedback and feedforward, are developed in the following section. The method of compensation is greatly dependent on the number of stages present (differential, second, or buffer stages).

In designing an op amp, one can begin at many points. The design procedure must be iterative, since it is almost impossible to relate all specifications simultaneously. For a typical CMOS op amp design, the following steps may be appropriate.

1. Decide on a Suitable Configuration

After examining the specifications in detail, determine the type of configuration required. For example, if extremely low noise and offset are a must, then a configuration that affords high gain in the input stage is required. If there are low-power requirements, then a Class AB-type output stage may be necessary. This in turn will govern the type of input stage that must be used. Often, one must create a configuration that meets a specific application.

2. Determine the Type of Compensation Needed to Meet the Specifications

There are many ways to compensate amplifiers. Some have unique aspects that make them suitable for particular configurations or specifications. For example, an op amp that must drive very large load capacitances might be compensated at the output. If this is the case, then

this requirement also dictates the types of input and output stages needed. As this example shows, iteration might be necessary between steps 1 and 2 of the design process.

3. Design Device Sizes for Proper dc, ac, and Transient Performance

This begins with hand calculations based on approximate design equations. Compensation components are also sized in this step of the procedure. After each device is sized by hand, a circuit simulator is used to fine-tune the design.

One may find during the design process that some specification may be difficult or impossible to meet with a given configuration. At this point the designer has to modify the configuration or search the literature for ideas particularly suited to the requirement. This literature search takes the place of creating a new configuration from scratch. For very critical designs, the hand calculations can achieve about 80% of the complete job in roughly 20% of the total job time. The remaining 20% of the job requires 80% of the time for completion. Sometimes hand calculations can be misleading due to their approximate nature. Nonetheless, they are necessary to give the designer a feel for the sensitivity of the design to parameter variation. There is no other way for the designer to understand how the various design parameters influence performance. Iteration by computer simulation gives the designer very little feeling for the design and is generally not a wise use of computer resources.*

In summary, the design process consists of two major steps. The first is the conception of the design and the second is the optimization of the design. The conception of the design is accomplished by proposing an architecture to meet the given specifications. This step is normally accomplished using hand calculations in order to maintain the intuitive viewpoint necessary for choices that must be made. The second step is to take the "first-cut" design and verify and optimize it. This is normally done by using computer simulation and can include such influences as environmental or process variations.

6.2 COMPENSATION OF OP AMPS

Operational amplifiers are generally used in a negative-feedback configuration. In this way, the relatively high, inaccurate forward gain can be used with feedback to achieve a very accurate transfer function that is a function of the feedback elements only. Figure 6.2-1 illustrates a general negative-feedback configuration. $A(s)$ is the amplifier gain and will normally be the open-loop, differential voltage gain of the op amp, and $F(s)$ is the transfer function for external feedback from the output of the op amp back to the input (see Section 5.6). The loop gain of this system will be defined as

$$\text{Loop gain} = L(s) = -A(s)F(s) \quad (6.2-1)$$

Consider a case where the forward gain from V_{in} to V_{out} is to be unity. It is easily shown that if the open-loop gain at dc $A(0)$ is between 1000 and 2000, and F is equal to 1, the forward gain varies from 0.999 to 0.9995. For very high loop gain (due primarily to a high amplifier gain),

*A useful rule in analog design is: (use of a simulator) $\times$ (common sense) = (a constant).

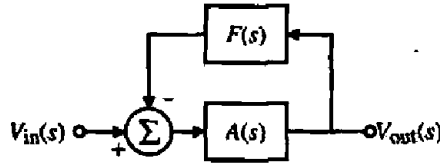


Figure 6.2-1 A single-loop, negative-feedback system.

the forward transfer function V_{out}/V_{in} is accurately controlled by the feedback network. This is the principle applied in using operational amplifiers.

Small-Signal Dynamics of a Two-Stage Op Amp

It is of primary importance that the signal fed back to the input of the op amp be of such amplitude and phase that it does not continue to regenerate itself around the loop. Should this occur, the result will be either clamping of the output of the amplifier at one of the supply potentials (regeneration at dc), or oscillation (regeneration at some frequency other than dc). The requirement for avoiding this situation can be succinctly stated by the following equation:

$$|A(j\omega_0)F(j\omega_0)| = |L(j\omega_0)| < 1 \quad (6.2-2)$$

where ω_0 is defined as

$$\text{Arg}[-A(j\omega_0)F(j\omega_0)] = \text{Arg}[L(j\omega_0)] = 0^\circ \quad (6.2-3)$$

Another convenient way to express this requirement is

$$\text{Arg}[-A(j\omega_{0\text{dB}})F(j\omega_{0\text{dB}})] = \text{Arg}[L(j\omega_{0\text{dB}})] > 0^\circ \quad (6.2-4)$$

where $\omega_{0\text{dB}}$ is defined as

$$|A(j\omega_{0\text{dB}})F(j\omega_{0\text{dB}})| = |L(j\omega_{0\text{dB}})| = 1 \quad (6.2-5)$$

If these conditions are met, the feedback system is said to be stable (i.e., sustained oscillation cannot occur).

This second relationship given in Eq. (6.2-4) is best illustrated with the use of Bode diagrams. Figure 6.2-2 shows the response of $|A(j\omega)F(j\omega)|$ and $\text{Arg}[-A(j\omega)F(j\omega)]$ as a function of frequency. The requirement for stability is that the $|A(j\omega)F(j\omega)|$ curve cross the 0 dB point before the $\text{Arg}[-A(j\omega)F(j\omega)]$ reaches 0° . A measure of stability is given by the value of the phase when $|A(j\omega)F(j\omega)|$ is unity, 0 dB. This measure is called *phase margin* and is described by the following relationship:

$$\text{Phase margin} = \Phi_M = \text{Arg}[-A(j\omega_{0\text{dB}})F(j\omega_{0\text{dB}})] = \text{Arg}[L(j\omega_{0\text{dB}})] \quad (6.2-6)$$

The importance of “good stability” obtained with adequate phase margin is best understood by considering the response of the closed-loop system in the time domain. Figure 6.2-3 shows the time response of a second-order closed-loop system with various phase margins. One can see that larger phase margins result in less “ringing” of the output signal. Too much

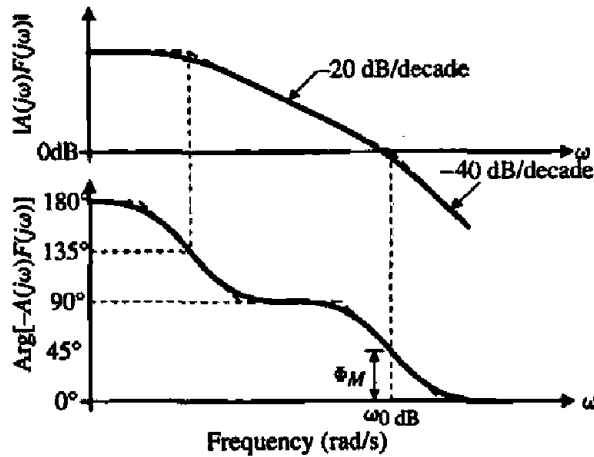


Figure 6.2-2 Frequency and phase response of a second-order system.

ringing can be undesirable, so it is important to have adequate phase margin keeping the ringing to an acceptable level. It is desirable to have a phase margin of at least 45° , with 60° preferable in most situations. Appendix C develops the relationship between phase margin and time domain response for second-order systems.

Now consider the second-order, small-signal model for an uncompensated op amp shown in Fig. 6.2-4. In order to generalize the results, the components associated with the first stage have the subscript *I* and those associated with the second stage have the subscript *II*. The locations for the two poles are given by the following equations:

$$p'_1 = \frac{-1}{R_f C_f} \quad (6.2-7)$$

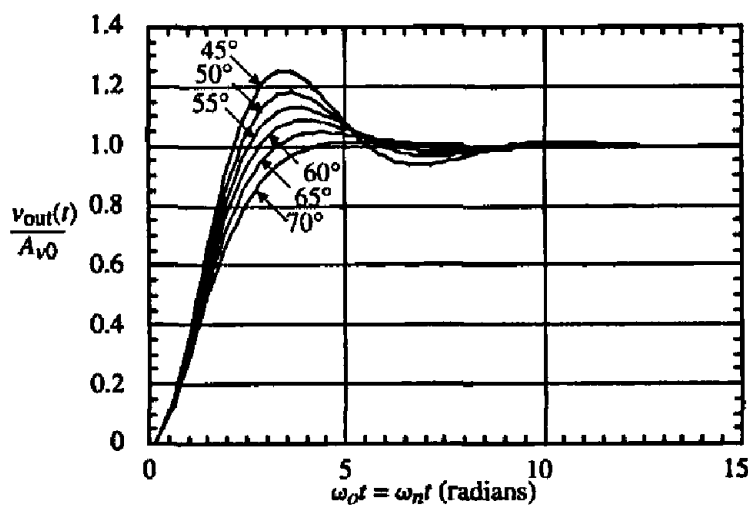


Figure 6.2-3 Response of a second-order system with various phase margins.

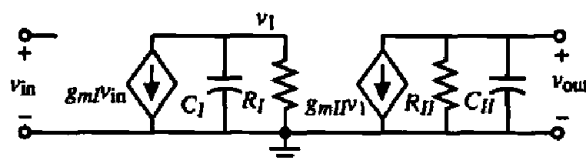


Figure 6.2-4 Second-order, small-signal equivalent circuit for a two-stage op amp.

and

$$p_2' = \frac{-1}{R_{II}C_{II}} \quad (6.2-8)$$

where R_I (R_{II}) is the resistance to ground seen from the output of the first (second) stage and C_I (C_{II}) is the capacitance to ground seen from the output of the first (second) stage. In a typical case, these poles are far away from the origin of the complex frequency plane and are relatively close together. Figure 6.2-5 illustrates the open-loop frequency response of a negative-feedback loop using the op amp modeled by Fig. 6.2-4 and a feedback factor of $F(s) = 1$. Note that $F(s) = 1$ is the worst case for stability considerations. In Fig. 6.2-5, the phase margin is significantly less than 45° , which means that the op amp should be compensated before using it in a closed-loop configuration.

Miller Compensation of the Two-Stage Op Amp

The first compensation method discussed here will be the "Miller" compensation technique [2]. This technique is applied by connecting a capacitor from the output to the input of the second transconductance stage g_{mII} . The resulting small-signal model is illustrated in Fig. 6.2-6. Two results come from adding the compensation capacitor C_c . First, the effective capacitance

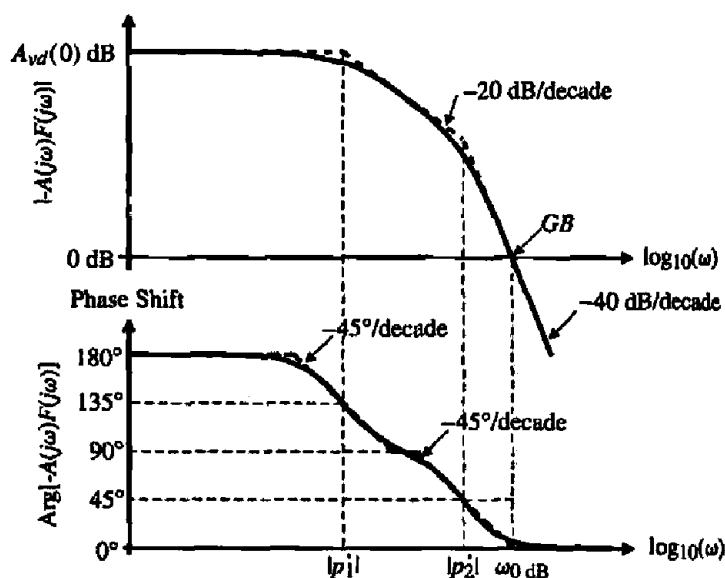


Figure 6.2-5 The open-loop frequency response of a negative-feedback loop using an uncompensated op amp and a feedback factor of $F(s) = 1$.

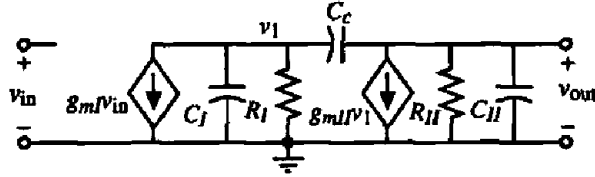


Figure 6.2-6 Miller capacitance applied to the two-stage op amp.

shunting R_I is increased by the additive amount of approximately $g_{mII}(R_{II})(C_c)$. This moves p_1 (the new location of p'_1) closer to the origin of complex frequency plane by a significant amount (assuming that the second-stage gain is large). Second, p_2 (the new location of p'_2) is moved away from the origin of the complex frequency plane, resulting from the negative feedback reducing the output resistance of the second stage.

The following derivation illustrates this in a rigorous way. The overall transfer function that results from the addition of C_c is

$$\frac{V_O(s)}{V_{in}(s)} = \frac{(g_{mI})(g_{mII})(R_I)(R_{II})(1 - sC_c/g_{mII})}{1 + s[R_I(C_I + C_c) + R_{II}/C_{II} + C_c] + s^2 R_I R_{II} [C_I C_{II} + C_c C_I + C_c C_{II}]} \quad (6.2-9)$$

Using the approach developed in Section 5.3 for two widely spaced poles gives the following compensated poles:

$$p_1 \cong \frac{-1}{g_{mII} R_I R_{II} C_c} \quad (6.2-10)$$

and

$$p_2 \cong \frac{-g_{mII} C_c}{C_I C_{II} + C_{II} C_c + C_I C_c} \quad (6.2-11)$$

If C_{II} is much greater than C_I and C_c is greater than C_b , then Eq. (6.2-11) can be approximated by

$$p_2 \cong \frac{-g_{mII}}{C_{II}} \quad (6.2-12)$$

It is of interest to note that a zero occurs on the positive-real axis of the complex frequency plane and is due to the feedforward path through C_c . The right half-plane zero is located at

$$z_1 = \frac{g_{mII}}{C_c} \quad (6.2-13)$$

Figure 6.2-7(a) illustrates the movement of the poles from their uncompensated to their compensated positions on the complex frequency plane. Figure 6.2-7(b) shows results of compensation illustrated by an asymptotic magnitude and phase plot. Note that the second pole does not begin to affect the magnitude until after $|A(j\omega)F(j\omega)|$ is less than unity. The right half-plane (RHP) zero increases the phase shift [acts like a left half-plane (LHP) pole] but

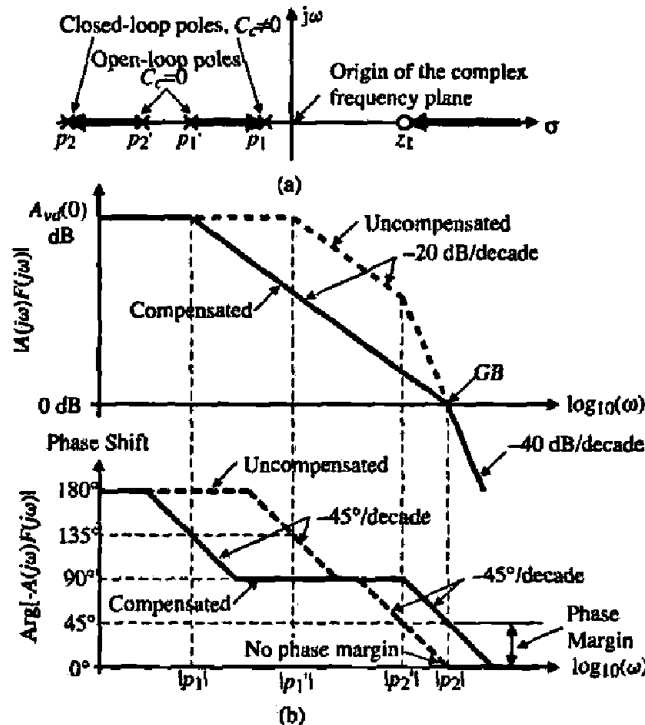


Figure 6.2-7 (a) Root locus plot of the loop gain $[F(s) = 1]$ resulting from the Miller compensation as C_c is varied from 0 to the value used to achieve the unprimed roots. (b) The asymptotic magnitude and phase plots of the loop gain $[F(s) = 1]$ before and after compensation.

increases the magnitude (acts like an LHP zero). Consequently, the RHP zero causes the two worst things possible with regard to stability considerations. If either the zero (z_1) or the pole (p_2) moves toward the origin of the complex frequency plane, the phase margin will be degraded. The task in compensating an amplifier for closed-loop applications is to move all poles and zeros, except for the dominant pole (p_1), sufficiently away from the origin of the complex frequency plane (beyond the unity-gain bandwidth frequency) to result in a phase shift similar to Fig. 6.2-7(b).

Only a second-order (two-pole) system has been considered thus far. In practice, there are more than two poles in the transfer function of a CMOS op amp. The rest of this treatment will concentrate on the two most dominant (smaller) poles and the RHP zero. Figure 6.2-8

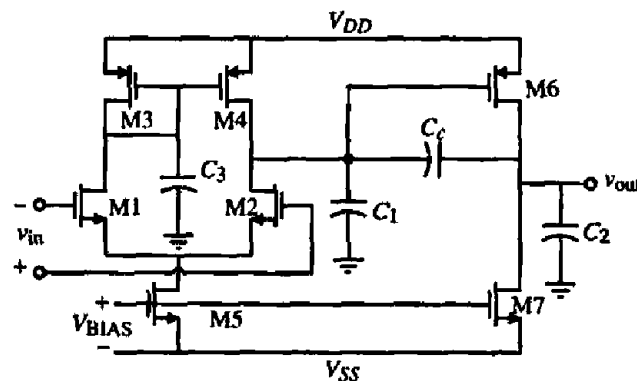


Figure 6.2-8 A two-stage op amp with various parasitic and circuit capacitances shown.

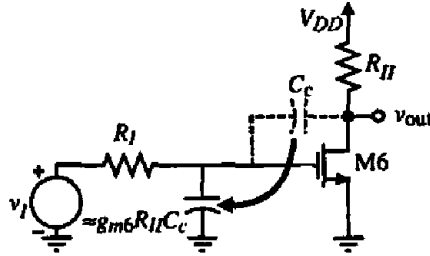


Figure 6.2-9 Illustration of the implementation of the dominant pole through the Miller effect on C_c . M6 is treated as an NMOS for this illustration.

illustrates a typical CMOS op amp with various parasitic and circuit capacitances shown. The approximate pole and zero locations resulting from these capacitances are given below:

$$p_1 \cong \frac{-G_I G_{II}}{g_{mI} C_c} = \frac{-(g_{ds2} + g_{ds4})(g_{ds6} + g_{ds7})}{g_{m6} C_c} \quad (6.2-14)$$

$$p_2 \cong \frac{-g_{mII}}{C_{II}} = \frac{-g_{m6}}{C_2} \quad (6.2-15)$$

and

$$z_1 \cong \frac{g_{mII}}{C_c} = \frac{g_{m6}}{C_c} \quad (6.2-16)$$

The unity-gain bandwidth as defined in Section 6.1 is easily derived (see Problem 6.2-3) and is shown to be approximately

$$GB \cong \frac{g_{mI}}{C_c} = \frac{g_{m2}}{C_c} \quad (6.2-17)$$

The above three roots are very important to the dynamic performance of the two-stage op amp. The dominant left-half plane pole, p_1 , is called the *Miller pole* and accomplishes the desired compensation. Intuitively, it is created by the Miller effect on the capacitance, C_c , as illustrated in Fig. 6.2-9, where M6 is assumed to be an NMOS transistor. The capacitor C_c is multiplied by approximately the gain of the second stage, $g_{II} R_{II}$, to give a capacitor in parallel with R_I of $g_{II} R_{II} C_c$. Multiplying this capacitance times R_I and inverting gives Eq. (6.2-14).

The second root of importance is p_2 . The magnitude of this root must be at least equal to GB and is due to the capacitance at the output of the op amp. It is often called the *output pole*. Generally, C_{II} is equal to the load capacitance, C_L , which makes the output pole strongly dependent on the load capacitance. Figure 6.2-10 shows intuitively how this root develops.

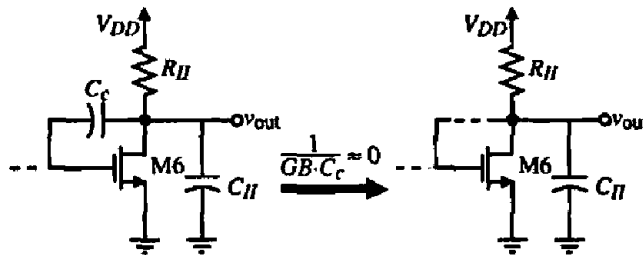


Figure 6.2-10 Illustration of the how the output pole in a two-stage op amp is created. M6 is treated as an NMOS for this illustration.

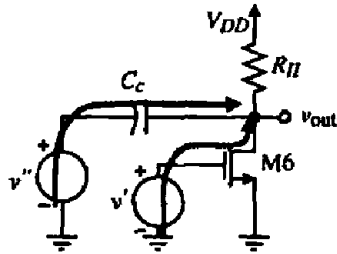


Figure 6.2-11 Illustration of how the RHP zero is developed. M6 is treated as an NMOS for this illustration.

Since $|p_2|$ is near or greater than GB , the reactance of C_c is approximately $1/(GB \cdot C_c)$ and is very small. For all practical purposes the drain of M6 is connected to the gate of M6, forming a MOS diode. We know that the small-signal resistance of a MOS diode is $1/g_m$. Multiplying $1/g_m$ by C_c (or C_L) and inverting gives Eq. (6.2-15).

The third root is the RHP zero. This is a very undesirable root because it boosts the loop gain magnitude while causing the loop phase shift to become more negative. Both of these results worsen the stability of the op amp. In BJT op amps, the RHP zero was not serious because of the large values of transconductance. However, in CMOS op amps, the RHP zero cannot be ignored. This zero comes from the fact that there are two signal paths from the input to the output as illustrated in Fig. 6.2-11. One path is from the gate of M6 through the compensation capacitor, C_c , to the output (V'' to V_{out}). The other path is through the transistor, M6, to the output (V' to V_{out}). At some complex frequency, the signals through these two paths will be equal and opposite and cancel, creating the zero. The RHP zero is developed by using superposition on these two paths as shown below:

$$V_{out}(s) = \left(\frac{-g_{m6} R_{II} (1/sC_c)}{R_{II} + 1/sC_c} \right) V' + \left(\frac{R_{II}}{R_{II} + 1/sC_c} \right) V'' = \frac{-R_{II} (g_{m6}/sC_c - 1)}{R_{II} + 1/sC_c} V \quad (6.2-18)$$

where $V = V' = V''$.

As stated before, the goal of the compensation task is to achieve a phase margin greater than 45° . It can be shown (see Problem 6.2-4) that if the zero is placed at least ten times higher than the GB , then in order to achieve 45° phase margin, the second pole (p_2) must be placed at least 1.22 times higher than GB . In order to obtain 60° of phase margin, p_2 must be placed about 2.2 times higher than GB as shown in the following example.

LOCATION OF THE OUTPUT POLE FOR A PHASE MARGIN OF 60°

For an op amp model with two poles and one RHP zero, prove that if the zero is ten times higher than GB , then in order to achieve a 60° phase margin, the second pole must be placed at least 2.2 times higher than GB .

Solution

The requirement for a 60° phase margin is given as

$$\Phi_M = \pm 180^\circ - \text{Arg}[A(j\omega)F(j\omega)] = \pm 180^\circ - \tan^{-1}\left(\frac{\omega}{|p_1|}\right) - \tan^{-1}\left(\frac{\omega}{|p_2|}\right) - \tan^{-1}\left(\frac{\omega}{z_1}\right) = 60^\circ$$

Assuming that the unity-gain frequency is GB , we replace ω by GB to get

$$120^\circ = \tan^{-1}\left(\frac{GB}{|p_1|}\right) + \tan^{-1}\left(\frac{GB}{|p_2|}\right) + \tan^{-1}\left(\frac{GB}{z_1}\right) = \tan^{-1}[A_v(0)] + \tan^{-1}\left(\frac{GB}{|p_2|}\right) + \tan^{-1}(0.1)$$

Assuming that $A_v(0)$ is large, then the above equation can be reduced to

$$24.3^\circ \approx \tan^{-1}\left(\frac{GB}{|p_2|}\right)$$

which gives $|p_2| \geq 2.2GB$.

Assuming that a 60° phase margin is required, the following relationships apply:

$$\frac{g_{m6}}{C_c} > 10 \left(\frac{g_{m2}}{C_c} \right) \quad (6.2-19)$$

Therefore,

$$g_{m6} > 10g_{m2} \quad (6.2-20)$$

Furthermore,

$$\left(\frac{g_{m6}}{C_c} \right) > 2.2 \left(\frac{g_{m2}}{C_c} \right) \quad (6.2-21)$$

Combining Eqs. (6.2-20) and (6.2-21) gives the following requirement:

$$C_c > \frac{2.2 C_2}{10} = 0.22 C_2 \quad (6.2-22)$$

Up to this point, we have neglected the influence of the capacitor, C_3 , associated with the current-mirror load of the input stage in Fig. 6.2-8. A small-signal model for the input stage of Fig. 6.2-8 that includes C_3 is shown in Fig. 6.2-12(a). The transfer function from the input to the output voltage of the first stage, $V_{o1}(s)$, can be written as

$$\begin{aligned} \frac{V_{o1}(s)}{V_{in}(s)} &= \frac{-g_{m1}}{2(g_{ds2} + g_{ds4})} \left[\frac{g_{m3} + g_{ds1} + g_{ds3}}{g_{m3} + g_{ds1} + g_{ds3} + sC_3} + 1 \right] \\ &\approx \frac{-g_{m1}}{2(g_{ds2} + g_{ds4})} \left[\frac{sC_3 + 2g_{m3}}{sC_3 + g_{m3}} \right] \end{aligned} \quad (6.2-23)$$

We see that there is a pole and a zero given as

$$p_3 = -\frac{g_{m3}}{C_3} \quad \text{and} \quad z_3 = -\frac{2g_{m3}}{C_3} \quad (6.2-24)$$

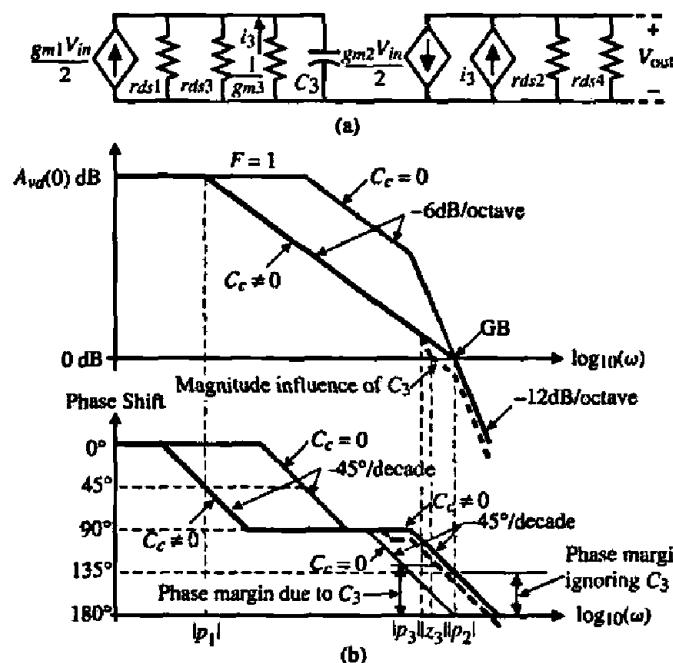


Figure 6.2-12 (a) Influence of the mirror pole, p_3 , on the Miller compensation of a two-stage op amp. (b) The location of the open-loop and closed-loop roots.

Fortunately, the presence of the zero tends to negate the effect of the pole. Generally, the pole and zero due to C_3 is greater than GB and will have very little influence on the stability of the two-stage op amp. Figure 6.2-12(b) illustrates the case where these roots are less than GB and even then they have little effect on stability. In fact, they actually increase the phase margin slightly because GB is decreased.

Controlling the Right Half-Plane Zero

The RHP zero resulting from the feedforward path through the compensation capacitor tends to limit the GB that might otherwise be achievable if the zero were not present. There are several ways of eliminating the effect of this zero. One approach is to eliminate the feedforward path by placing a unity-gain buffer in the feedback path of the compensation capacitor [3]. This technique is shown in Fig. 6.2-13. Assuming that the output resistance of the

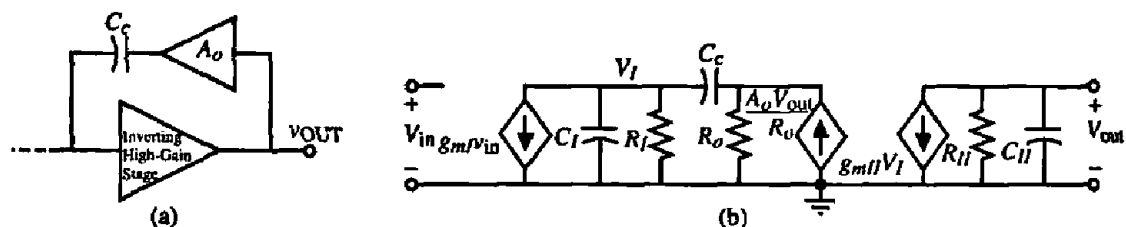


Figure 6.2-13 (a) Elimination of the feedforward path by a voltage amplifier. (b) Small-signal model for the two-stage op amp using the technique of (a).

unity-gain buffer is small ($R_o \rightarrow 0$), then the transfer function is given by the following equation:

$$\frac{V_o(s)}{V_{in}(s)} = \frac{(g_{mI})(g_{mII})(R_I)(R_{II})}{1 + s[R_I C_I + R_{II} C_{II} + R_I C_c + g_{mII} R_I R_{II} C_c] + s^2[R_I R_{II} C_{II} (C_I + C_c)]} \quad (6.2-25)$$

Using the technique as before to approximate p_1 and p_2 results in the following:

$$p_1 \cong \frac{-1}{R_I C_I + R_{II} C_{II} + R_I C_c + g_{mII} R_I R_{II} C_c} \cong \frac{-1}{g_{mII} R_I R_{II} C_c} \quad (6.2-26)$$

and

$$p_2 \cong \frac{-g_{mII} C_c}{C_{II} (C_I + C_c)} \quad (6.2-27)$$

Note that the poles of the circuit in Fig. 6.2-13 are approximately the same as before, but the zero has been removed. With the zero removed, the pole p_2 can be placed higher than the GB in order to achieve a phase margin of 45° . For a 60° phase margin, p_2 must be placed 1.73 times greater than the GB . Using the type of compensation scheme shown in Fig. 6.2-13 results in greater bandwidth capabilities as a result of eliminating the zero.

The above analysis neglects the output resistance of the buffer amplifier R_o , which can be significant. Taking the output resistance into account, and assuming that it is less than R_I or R_{II} , results in an additional pole p_4 and an LHP zero z_2 given by

$$p_4 \cong \frac{-1}{R_o [C_I C_c / (C_I + C_c)]} \quad (6.2-28)$$

$$z_2 \cong \frac{-1}{R_o C_c} \quad (6.2-29)$$

Although the LHP zero can be used for compensation, the additional pole makes this method less desirable than the following method. The most important aspect of this result is that it leads naturally to the next scheme for controlling the RHP zero.

Another means of eliminating the effect of the RHP zero resulting from feedforward through the compensation capacitor C_c is to insert a nulling resistor in series with C_c [4]. Figure 6.2-14 shows the application of this technique. This circuit has the following node-voltage equations:

$$g_{mI} V_{in} + \frac{V_I}{R_I} + s C_I V_I + \left(\frac{s C_c}{1 + s C_c R_z} \right) (V_I - V_{out}) = 0 \quad (6.2-30)$$

$$g_{mII} V_I + \frac{V_o}{R_{II}} + s C_{II} V_{out} + \left(\frac{s C_c}{1 + s C_c R_z} \right) (V_{out} - V_I) = 0 \quad (6.2-31)$$

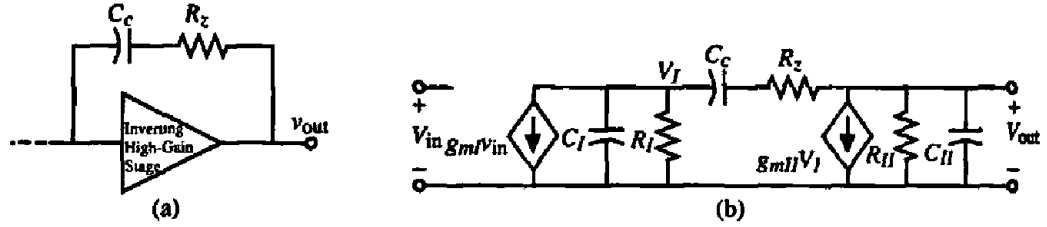


Figure 6.2-14 (a) Use of a nulling resistor, R_z , to control the RHP zero. (b) Small-signal model of the nulling resistor applied to a two-stage op amp.

These equations can be solved to give

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{a\{1 - s[(C_c/g_{m2}) - R_z C_c]\}}{1 + bs + cs^2 + ds^3} \quad (6.2-32)$$

where

$$a = g_{m1}g_{m2}R_I R_{II} \quad (6.2-33)$$

$$b = (C_{II} + C_c)R_{II} + (C_I + C_c)R_I + g_{m2}R_I R_{II} C_c + R_z C_c \quad (6.2-34)$$

$$c = [R_I R_{II}(C_I C_{II} + C_c C_I + C_c C_{II}) + R_z C_c (R_I C_I + R_{II} C_{II})] \quad (6.2-35)$$

$$d = R_I R_{II} R_z C_I C_{II} C_c \quad (6.2-36)$$

If R_z is assumed to be less than R_I or R_{II} and the poles are widely spaced, then the roots of Eq. (6.2-32) can be approximated as

$$p_1 \cong \frac{-1}{(1 + g_{m2}R_{II})R_I C_c} \cong \frac{-1}{g_{m2}R_{II}R_I C_c} \quad (6.2-37)$$

$$p_2 \cong \frac{-g_{m2}C_c}{C_I C_{II} + C_c C_I + C_c C_{II}} \cong \frac{-g_{m2}}{C_{II}} \quad (6.2-38)$$

$$p_4 = \frac{-1}{R_z C_I} \quad (6.2-39)$$

and

$$z_1 = \frac{1}{C_c(1/g_{m2} - R_z)} \quad (6.2-40)$$

It is easy to see how the nulling resistor accomplishes the control over the RHP zero. Figure 6.2-15 shows the output stage broken into two parts, similar to what was done in Fig. 6.2-11. The output voltage, V_{out} , can be written as

$$V_{out} = \frac{-g_{m6}R_{II}\left(R_z + \frac{1}{sC_c}\right)}{R_{II} + R_z + \frac{1}{sC_c}} V' + \frac{R_{II}}{R_{II} + R_z + \frac{1}{sC_c}} V'' = \frac{-R_{II}\left(g_{m6}R_z + \frac{g_{m6}}{sC_c} - 1\right)}{R_{II} + R_z + \frac{1}{sC_c}} \quad (6.2-41)$$

Setting the numerator equal to zero gives Eq. (6.2-40), assuming $g_{m6} = g_{mII}$.

The resistor R_z allows independent control over the placement of the zero. In order to remove the RHP zero, R_z must be set equal to $1/g_{mII}$. Another option is to move the zero from the RHP to the LHP and place it on top of p_2 . As a result, the pole associated with the output loading capacitance is canceled. To accomplish this, the following condition must be satisfied:

$$z_1 = p_2 \quad (6.2-42)$$

which results in

$$\frac{1}{C_c(1/g_{mII} - R_z)} = \frac{-g_{mII}}{C_{II}} \quad (6.2-43)$$

The value of R_z can be found as

$$R_z = \left(\frac{C_c + C_{II}}{C_c}\right)\left(\frac{1}{g_{mII}}\right) \quad (6.2-44)$$

With p_2 canceled, the remaining roots are p_1 and p_3 . For unity-gain stability, all that is required is that the magnitudes of p_3 and p_4 be sufficiently greater than GB . Therefore

$$|p_3| > A_v(0)|p_1| = \frac{A_v(0)}{g_{mII}R_{II}R_I C_c} = GB \quad (6.2-45)$$

$$|p_4| = (1/R_z C_I) > (g_{mI}/C_c) \quad (6.2-46)$$

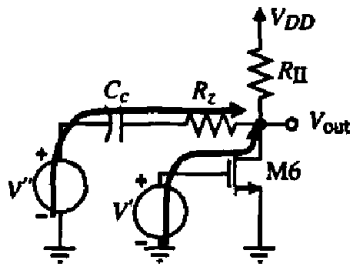


Figure 6.2-15 Illustration of how the nulling resistor accomplishes the control of the RHP zero.

Substituting Eq. (6.2-44) into Eq. (6.2-46) and assuming $C_H \gg C_c$ results in

$$C_c > \sqrt{\frac{g_{m1}}{g_{m11}}} C_1 C_H \quad (6.2-47)$$

The nulling resistor approach has been used in the two-stage op amp with excellent results. The op amp can have good stability properties even with a large load capacitor. The only drawback is that the output pole, p_2 , cannot change after the compensation has been designed (as it will if C_L changes).

The magnitude of the output pole, p_2 , can be increased by introducing gain in the Miller capacitor feedback path as done by M8 in Fig. 6.2-16(a) [5]. The small-signal model of Fig. 6.2-16(a) is shown in Fig. 6.2-16(b). The resistors R_1 and R_2 are defined as

$$R_1 = \frac{1}{g_{ds2} + g_{ds4} + g_{ds9}} \quad (6.2-48)$$

and

$$R_2 = \frac{1}{g_{ds6} + g_{ds7}} \quad (6.2-49)$$

where transistors M2 and M4 are the output transistors of the first stage. In order to simplify the analysis, we have rearranged the controlled source, $g_{m8}V_{s8}$, and have ignored r_{ds8} . The

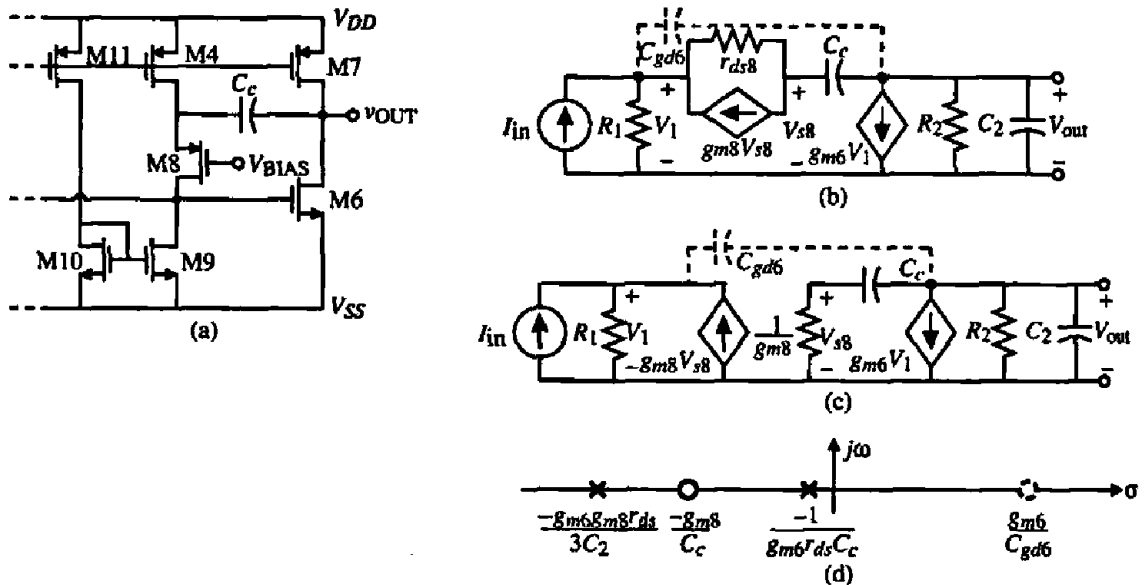


Figure 6.2-16 (a) Circuit to increase the magnitude of the output pole. (b) Small-signal model of (a). (c) Simplified version of (b). (d) Resulting poles and zeros of (a).

simplified small-signal model of Fig. 6.2-16(a) is given in Fig. 6.2-16(c). The nodal equations of this circuit are

$$I_{in} = G_1 V_1 - g_{m8} V_{s8} = G_1 V_1 - \left(\frac{g_{m8} s C_c}{g_{m8} + s C_c} \right) V_{out} \quad (6.2-50)$$

and

$$0 = g_{m6} V_1 + \left(G_2 + s C_2 + \frac{g_{m6} s C_c}{g_{m8} + s C_c} \right) V_{out} \quad (6.2-51)$$

Solving for the transfer function V_{out}/I_{in} gives

$$\frac{V_{out}}{I_{in}} = \left(\frac{-g_{m6}}{G_1 G_2} \right) \left[\frac{\left(1 + \frac{s C_c}{g_{m8}} \right)}{1 + s \left(\frac{C_c}{g_{m8}} + \frac{C_2}{G_2} + \frac{C_c}{G_2} + \frac{g_{m6} C_c}{G_1 G_2} \right) + s^2 \left(\frac{C_c C_2}{g_{m8} G_2} \right)} \right] \quad (6.2-52)$$

Using the approximate method of solving for the roots of the denominator illustrated earlier gives

$$p_1 = \frac{-1}{\frac{C_c}{g_{m8}} + \frac{C_c}{G_2} + \frac{C_2}{G_2} + \frac{g_{m6} C_c}{G_1 G_2}} \approx \frac{-6}{g_{m6} r_{ds}^2 C_c} \quad (6.2-53)$$

and

$$p_2 \approx \frac{\frac{g_{m6} r_{ds}^2 C_c}{6}}{\frac{C_c C_2}{g_{m8} G_2}} = \frac{g_{m8} r_{ds}^2 G_2}{6} \left(\frac{g_{m6}}{C_2} \right) = \left(\frac{g_{m8} r_{ds}}{3} \right) |p'_2| \quad (6.2-54)$$

where all the various channel resistances have been assumed to equal r_{ds} and p'_2 is the output pole for normal Miller compensation. The result of Fig. 6.2-16(a) is to keep the dominant pole approximately the same and to multiply the output pole by roughly the gain of a single stage ($g_{m8} r_{ds}$). In addition to the poles, there is an LHP zero at $g_{m6}/s C_c$, which can be used to enhance the compensation. Note in Figure 6.2-16(d) that there is still an RHP zero in the compensation scheme of Fig. 6.2-16 because of the feedforward path through C_{gs6} shown as dashed path on Figs. 6.2-16(b) and 6.2-16(c).

Figure 6.2-17 shows intuitively how the output pole is increased by the addition of M8 in the feedback path around M6. At frequencies near GB , the reactance of C_c can be considered small. Under these assumptions, M6 approximates an MOS diode with a gain of $g_{m6} r_{ds6}$ in the feedback path. This makes the output resistance seen by C_2 (R_{II}) to be approximately

$$R_{out} = R_{II} \parallel \left(\frac{1}{g_{m6} g_{m8} r_{ds8}} \right) \approx \frac{1}{g_{m6} g_{m8} r_{ds8}} \quad (6.2-55)$$

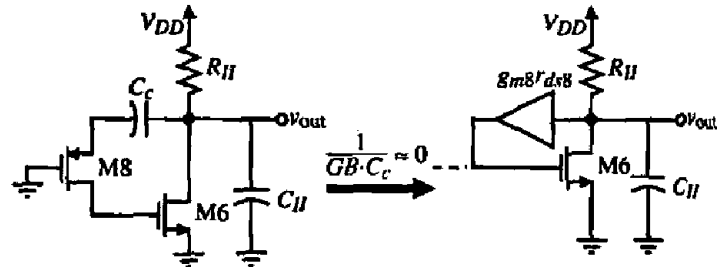


Figure 6.2-17 Illustration of how the output pole is increased by Fig. 6.2-16(a).

Multiplying this resistance by C_2 and inverting gives Eq. (6.2-54), ignoring the influence of the channel resistances of other transistors, which leads to the 3 in the denominator.

Feedforward Compensation

Another compensation technique used in CMOS op amps is the feedforward scheme shown in Fig. 6.2-18(a). In this circuit, the buffer is used to break the bidirectional path through the compensation capacitor. Unfortunately, this circuit will result in a zero that is in the right half-plane. If either the polarity of the buffer or the high-gain amplifier is reversed, the zero will be in the left half-plane. Figure 6.2-18(b) shows a feedforward compensation technique that has a zero in the left half-plane because the gain of the buffer is inverted. Figure 6.2-18(c) can be used as a model for this circuit. The voltage-transfer function $V_{out}(s)/V_{in}(s)$ can be found to be

$$\frac{V_{out}(s)}{V(s)} = \frac{AC_c}{C_c + C_{II}} \left(\frac{s + g_{mII}/AC_c}{s + 1/[R_{II}(C_c + C_{II})]} \right) \quad (6.2-56)$$

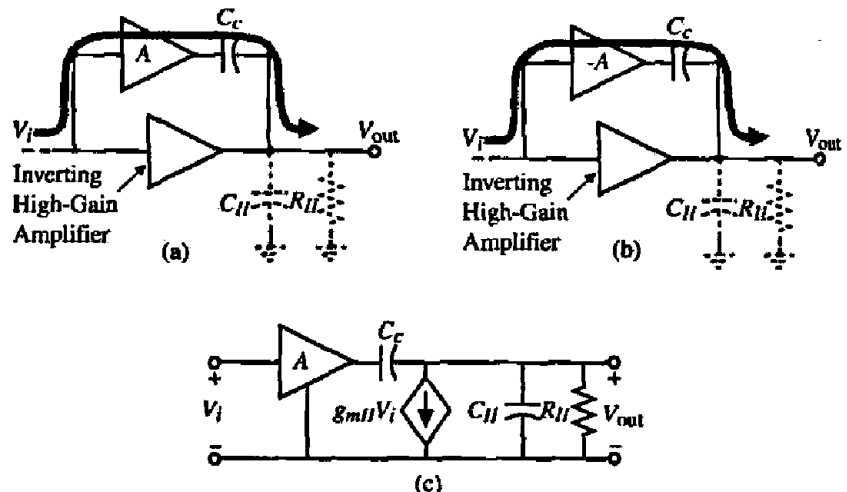


Figure 6.2-18 (a) Feedforward resulting in an RHP zero. (b) Feedforward resulting in an LHP zero. (c) Small-signal model for (b).

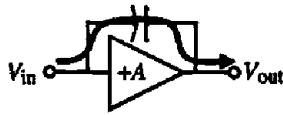


Figure 6.2-19 Feedforward compensation around a noninverting amplifier.

In order to use the circuit in Fig. 6.2-18(b) to achieve compensation, it is necessary to place the zero located at g_{mf}/AC_c above the value of GB so that the boosting of the magnitude will not negate the desired effect of positive phase shift caused by the zero. Fortunately, the phase effects extend over a much broader frequency range than the magnitude effects so that this method will contribute additional phase margin to that provided by the feedback compensation technique. It is quite possible that several zeros can be generated in the transfer function of the op amp. These zeros should all be placed above GB and should be well controlled to avoid large settling times caused by poles and zeros that are close together in the transient response [6].

Another form of feedforward compensation is to provide a feedforward path around a noninverting amplifier. This is shown in Fig. 6.2-19. This type of compensation is often used in source followers, where a capacitor is connected from the gate to source of the source follower. The capacitor will provide a path that bypasses the transistor at high frequencies.

6.3 DESIGN OF THE TWO-STAGE OP AMP

The previous two sections described the general approach to op amp design and compensation. In this section, a procedure will be developed that will enable a first-cut design of the two-stage op amp shown in Fig. 6.3-1. In order to simplify the notation, it is convenient to define the notation $S_i = W_i/L_i = (W/L)_i$, where S_i is the ratio of W and L of the i th transistor.

Design Procedure for the Two-stage CMOS Op Amp

Before beginning this task, important relationships describing op amp performance will be summarized from Section 6.2, assuming that $g_{m1} = g_{m2} = g_{m1}$, $g_{m6} = g_{m1}$, $g_{ds2} + g_{ds4} = G_1$, and $g_{ds6} + g_{ds7} = G_H$. These relationships are based on the circuit shown in Fig. 6.3-1.

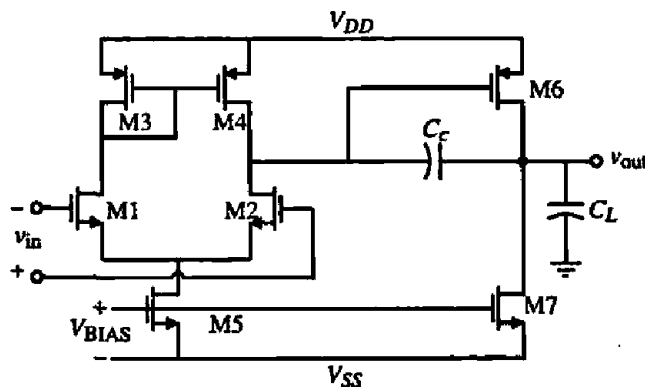


Figure 6.3-1 Schematic of an unbuffered, two-stage CMOS op amp with an n-channel input pair.

$$\text{Slew rate } SR = \frac{I_5}{C_c} \quad (6.3-1)$$

$$\text{First-stage gain } A_{v1} = \frac{-g_{m1}}{g_{ds2} + g_{ds4}} = \frac{-2g_{m1}}{I_5(\lambda_2 + \lambda_4)} \quad (6.3-2)$$

$$\text{Second-stage gain } A_{v2} = \frac{-g_{m6}}{g_{ds6} + g_{ds7}} = \frac{-g_{m6}}{I_6(\lambda_6 + \lambda_7)} \quad (6.3-3)$$

$$\text{Gain bandwidth } GB = \frac{g_{m1}}{C_c} \quad (6.3-4)$$

$$\text{Output pole } p_2 = \frac{-g_{m6}}{C_L} \quad (6.3-5)$$

$$\text{RHP zero } z_1 = \frac{g_{m6}}{C_c} \quad (6.3-6)$$

$$\text{Positive CMR } V_{in}(\max) = V_{DD} - \sqrt{\frac{I_5}{\beta_3}} - |V_{T03}|(\max) + V_{T1}(\min) \quad (6.3-7)$$

$$\text{Negative CMR } V_{in}(\min) = V_{SS} + \sqrt{\frac{I_5}{\beta_1}} + V_{T1}(\max) + V_{DS5}(\text{sat}) \quad (6.3-8)$$

$$\text{Saturation voltage } V_{DS}(\text{sat}) = \sqrt{\frac{2I_{DS}}{\beta}} \quad (6.3-9)$$

It is assumed that all transistors are in saturation for the above relationships.

The following design procedure assumes that specifications for the following parameters are given:

1. Gain at dc, $A_v(0)$
2. Gain bandwidth, GB
3. Input common-mode range, ICMR
4. Load capacitance, C_L
5. Slew rate, SR
6. Output voltage swing
7. Power dissipation, P_{diss}

The design procedure begins by choosing a device length to be used throughout the circuit. This value will determine the value of the channel length modulation parameter λ , which will be a necessary parameter in the calculation of amplifier gain. Because transistor modeling varies strongly with channel length, the selection of a device length to be used in the design (where possible) allows for more accurate simulation models. Having chosen the nominal transistor device length, one next establishes the minimum value for the compensation

capacitor C_c . It was shown in Section 6.2 that placing the output pole p_2 2.2 times higher than the GB permitted a 60° phase margin (assuming that the RHP zero z_1 is placed at or beyond ten times GB). It was shown in Eq. (6.2-22) that such pole and zero placements result in the following requirement for the minimum value for C_c :

$$C_c > (2.2/10)C_L \quad (6.3-10)$$

Next, determine the minimum value for the tail current I_5 , based on slew-rate requirements. Using Eq. (6.3-1), the value for I_5 is determined to be

$$I_5 = SR(C_c) \quad (6.3-11)$$

If the slew-rate specification is not given, then one can choose a value based on settling-time requirements. Determine a value that is roughly ten times faster than the settling-time specification, assuming that the output slews approximately one-half of the supply rail. The value of I_5 resulting from this calculation can be changed later if need be. The aspect ratio of M3 can now be determined by using the requirement for positive input common-mode range. The following design equation for $(W/L)_3$ was derived from Eq. (6.3-7):

$$S_3 = (W/L)_3 = \frac{I_5}{(K'_3)[V_{DD} - V_{in}(\max) - |V_{T03}|(\max) + V_{T1}(\min)]^2} \quad (6.3-12)$$

If the value determined for $(W/L)_3$ is less than one, then it should be increased to a value that minimizes the product of W and L . This minimizes the area of the gate region, which in turn reduces the gate capacitance. This gate capacitance contributes to the mirror pole which may cause a degradation in phase margin.

Requirements for the transconductance of the input transistors can be determined from knowledge of C_c and GB . The transconductance g_{m1} can be calculated using the following equation:

$$g_{m1} = GB(C_c) \quad (6.3-13)$$

The aspect ratio $(W/L)_1$ is directly obtainable from g_{m1} as shown below:

$$S_1 = (W/L)_1 = \frac{g_{m1}^2}{(K'_1)(I_5)} \quad (6.3-14)$$

Enough information is now available to calculate the saturation voltage of transistor M5. Using the negative ICMR equation, calculate V_{DSS} using the following relationship derived from Eq. (6.3-8):

$$V_{DSS} = V_{in}(\min) - V_{SS} - \left(\frac{I_5}{\beta_1}\right)^{1/2} - V_{T1}(\max) \quad (6.3-15)$$

If the value for V_{DSS} is less than about 100 mV, then the possibility of a rather large $(W/L)_5$ may result. This may not be acceptable. If the value for V_{DSS} is less than zero, then the ICMR

specification may be too stringent. To solve this problem, I_3 can be reduced or $(W/L)_1$ increased. The effects of these changes must be accounted for in previous design steps. One must iterate until the desired result is achieved. With V_{DS3} determined, $(W/L)_3$ can be extracted using Eq. (6.3-9) in the following way:

$$S_3 = (W/L)_3 = \frac{2(I_3)}{K'_3(V_{DS3})^2} \quad (6.3-16)$$

At this point, the design of the first stage of the op amp is complete. We next consider the output stage.

For a phase margin of 60° , the location of the output pole was assumed to be placed at 2.2 times GB . Based on this assumption and the relationship for $|p_2|$ in Eq. (6.3-5), the transconductance g_{m6} can be determined using the following relationship:

$$g_{m6} = 2.2(g_{m2})(C_L/C_c) \quad (6.3-17)$$

Generally, for reasonable phase margin, the value of g_{m6} is approximately ten times the input stage transconductance g_{m1} . At this point, there are two possible approaches to completing the design of M6 (i.e., W/L_6 and I_6). The first is to achieve proper mirroring of the first-stage current-mirror load of Fig. 6.3-1 (M3 and M4). This requires that $V_{SG4} = V_{SG6}$. Using the formula for g_m , which is $K'S(V_{GS} - V_T)$, we can write that if $V_{SG4} = V_{SG6}$, then

$$S_6 = S_4 \frac{g_{m6}}{g_{m4}} \quad (6.3-18)$$

Knowing g_{m6} and S_6 will define the dc current I_6 using the following equation:

$$I_6 = \frac{g_{m6}^2}{(2)(K'_6)(W/L)_6} = \frac{g_{m6}^2}{2K'_6 S_6} \quad (6.3-19)$$

One must now check to make sure that the maximum output voltage specification is satisfied. If this is not true, then the current or W/L ratio can be increased to achieve a smaller $V_{DS}(\text{sat})$. If these changes are made to satisfy the maximum output voltage specification, then the proper current mirroring of M3 and M4 is no longer guaranteed.

The second approach to designing the output stage is to use the value of g_{m6} and the required $V_{DS}(\text{sat})$ of M6 to find the current. Combining the defining equation for g_m and $V_{DS}(\text{sat})$ results in an equation relating (W/L) , $V_{DS}(\text{sat})$, g_m , and process parameters. Using this relationship, given below, with the $V_{DS}(\text{sat})$ requirement taken from the output range specification one can determine $(W/L)_6$.

$$S_6 = (W/L)_6 = \frac{g_{m6}}{K'_6 V_{DS6}(\text{sat})} \quad (6.3-20)$$

Equation (6.3-19) is used as before to determine a value for I_6 . In either approach to finding I_6 , one also should check the power dissipation requirements since I_6 will most likely determine the majority of the power dissipation.

The device size of M7 can be determined from the balance equation given below:

$$S_7 = (W/L)_7 = (W/L)_5 \left(\frac{I_6}{I_5} \right) = S_5 \left(\frac{I_6}{I_5} \right) \quad (6.3-21)$$

The first-cut design of all W/L ratios is now complete. Figure 6.3-2 illustrates the above design procedure showing the various design relationships and where they apply in the two-stage CMOS op amp.

At this point in the design procedure, the total amplifier gain must be checked against the specifications.

$$A_v = \frac{(2)(g_{m2})(g_{m6})}{I_5(\lambda_2 + \lambda_4)I_6(\lambda_6 + \lambda_7)} \quad (6.3-22)$$

If the gain is too low, a number of things can be adjusted. The best way to do this is to use Table 6.3-1, which shows the effects of various device sizes and currents on the different parameters generally specified. Each adjustment may require another pass through this design procedure in order to ensure that all specifications have been met. Table 6.3-2 summarizes the above design procedure.

No attempt has been made to account for noise or PSRR thus far in the design procedure. Now that the preliminary design is complete, these two specifications can be addressed. The input-referred noise voltage results primarily from the load and input transistors of the first stage. Each of these contribute both thermal and $1/f$ noise. The $1/f$ noise contributed by any transistor can be reduced by increasing device area (e.g., increase WL). Thermal noise contributed by any transistor can be reduced by increasing its g_m . This is accomplished by an increase in W/L , an increase in current, or both. Effective input-noise voltage attributed to the load transistors can be reduced by reducing the g_{m3}/g_{m1} (g_{m4}/g_{m2}) ratio. One must be careful

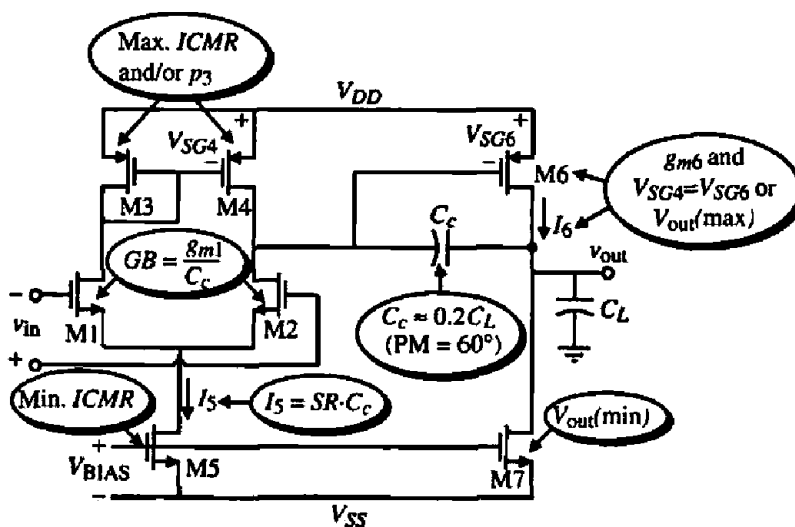


Figure 6.3-2 Illustration of the design relationship and the circuit for a two-stage CMOS op amp.

TABLE 6.3-1 Dependence of the Performance of Fig. 6.3-1 on dc Current, W/L Ratios, and the Compensating Capacitor

	Drain Current		M1 and M2		M3 and M4		Inverter	Inverter Load		Compensation Capacitor
	I_S	I_T	W/L	L	W	L	W_6/L_6	W_7	L_7	C_c
Increase dc Gain	$(\downarrow)^{1/2}$	$(\downarrow)^{1/2}$	$(\uparrow)^{1/2}$	$\uparrow$		$\uparrow$	$(\uparrow)^{1/2}$		$\uparrow$	$\downarrow$
Increase GB	$(\uparrow)^{1/2}$		$(\uparrow)^{1/2}$							$\downarrow$
Increase RHP Zero		$(\uparrow)^{1/2}$					$(\uparrow)^{1/2}$			$\downarrow$
Increase Slew Rate	$\uparrow$									$\downarrow$
Increase C_L										$\downarrow$

TABLE 6.3-2 Unbuffered Op Amp Design Procedure

This design procedure assumes that the gain at dc (A_v), unity-gain bandwidth (GB), input common-mode range [$V_{in}(\min)$ and $V_{in}(\max)$], load capacitance (C_L), slew rate (SR), settling time (T_s), output voltage swing [$V_{out}(\max)$ and $V_{out}(\min)$], and power dissipation (P_{dis}) are given.

1. Choose the smallest device length that will keep the channel modulation parameter constant and give good matching for current mirrors.
2. From the desired phase margin, choose the minimum value for C_c ; that is, for a 60° phase margin we use the following relationship. This assumes that $z \geq 10GB$.

$$C_c > 0.22C_L$$

3. Determine the minimum value for the "tail current" (I_S) from the largest of the two values.

$$I_S = SR \cdot C_c$$

$$I_S \cong 10 \left(\frac{V_{DD} + |V_{SS}|}{2T_s} \right)$$

4. Design for S_3 from the maximum input voltage specification.

$$S_3 = \frac{2I_S}{K'_3[V_{DD} - V_{in}(\max) - |V_{T03}|(\max) + V_{T1}(\min)]^2} \cong 1$$

5. Verify that the pole and zero due to C_{p3} and C_{n4} ($= 0.67W_3L_3C_{ox}$) will not be dominant by assuming p_3 to be greater than $10GB$.

$$\frac{g_{m3}}{2C_{p3}} > 10GB$$

6. Design for S_1 (S_2) to achieve the desired GB .

$$g_{m1} = GB \cdot C_c \Rightarrow S_1 = S_2 = \frac{g_{m1}}{K'_1 I_S}$$

TABLE 6.3-2 (continued)

7. Design for S_3 from the minimum input voltage. First calculate $V_{DSS}(\text{sat})$ then find S_3 .

$$V_{DSS}(\text{sat}) = V_{in}(\text{min}) - V_{SS} - \sqrt{\frac{I_3}{\beta_1}} - V_{T1}(\text{max}) \geq 100 \text{ mV}$$

$$S_3 = \frac{2I_3}{K'_3[V_{DSS}(\text{sat})]^2}$$

8. Find S_6 and I_6 by letting the second pole (p_2) be equal to 2.2 times GB .

$$g_{m6} = 2.2g_{m2}(C_L/C_c)$$

Let $V_{S04} = V_{S06}$, which gives

$$S_6 = S_4 \frac{g_{m6}}{g_{m4}}$$

Knowing g_{m6} and S_6 allows us to solve for I_6 as

$$I_6 = \frac{g_{m6}^2}{2K'_6 S_6}$$

9. Alternately, I_6 can be calculated by solving for S_6 using

$$S_6 = \frac{g_{m6}}{K'_6 V_{DSS6}(\text{sat})}$$

and then using the previous relationship to find I_6 . Of course, the proper mirror between M3 and M4 is no longer guaranteed.

10. Design S_7 to achieve the desired current ratios between I_5 and I_6 .

$$S_7 = (I_6/I_5)S_5$$

11. Check gain and power dissipation specifications.

$$A_v = \frac{2g_{m2}g_{m6}}{I_5(\lambda_2 + \lambda_3)(\lambda_6 + \lambda_7)}$$

$$P_{diss} = (I_5 + I_6)(V_{DD} + |V_{SS}|)$$

12. If the gain specification is not met, then the currents I_5 and I_6 can be decreased or the W/L ratios of M2 and/or M6 increased. The previous calculations must be rechecked to ensure that they have been satisfied. If the power dissipation is too high, then one can only reduce the currents I_5 and I_6 . Reduction of currents will probably necessitate an increase of some of the W/L ratios to satisfy input and output swings.

13. Simulate the circuit to check to see that all specifications are met.